

## Data Collection and Pre-processing Phase

| Field         | Details                         |
|---------------|---------------------------------|
| Date          | 20 August 2026                  |
| Team ID       | Individual                      |
| Project Title | Global AI Adoption in Education |
| Maximum Marks | 10 Marks                        |

## Data Exploration and Preprocessing

| Section                             | Description                                                                                                                                                                                                                                                                                              |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Overview</b>                | The dataset contains information related to <b>AI adoption in education</b> , including AI usage among students and teachers, school adoption, regional differences, rural and urban adoption, popular AI tools, and AI adoption trends. The dataset is used to create an interactive Tableau dashboard. |
| <b>Data Cleaning</b>                | Checked the dataset for <b>missing values, duplicate records, blank fields, and inconsistent values</b> . Unnecessary records and errors were removed or corrected to improve data quality.                                                                                                              |
| <b>Data Transformation</b>          | Data was <b>filtered and sorted</b> based on relevant fields. Required fields were prepared for Tableau visualization, and calculated fields were created where necessary for analysis.                                                                                                                  |
| <b>Data Type Conversion</b>         | Data types were checked and corrected where required. <b>Numerical fields</b> were converted to appropriate number formats, <b>categorical fields</b> were treated as text, and <b>date fields</b> were converted to date format where applicable.                                                       |
| <b>Column Splitting and Merging</b> | Columns were reviewed to determine whether any fields needed to be split or combined. Required fields were organized to make them suitable for Tableau analysis and visualization.                                                                                                                       |
| <b>Data Modeling</b>                | The dataset was structured for Tableau analysis. Relevant dimensions and measures were identified and used to create <b>KPIs, calculated fields, filters, and visualizations</b> .                                                                                                                       |
| <b>Save Processed Data</b>          | The cleaned and processed data was saved and connected to the <b>Tableau workbook (.twbx)</b> for dashboard development and future analysis.                                                                                                                                                             |

## Dataset Attributes

The dataset includes the following fields:

- Year
- Month
- Country
- Region
- Student AI Usage (%)
- Teacher AI Usage (%)
- Schools AI Adoption (%)
- Average Daily AI Usage (Minutes)
- Top AI Tool
- Internet Penetration (%)
- Education Index
- Government AI Policy
- Urban AI Usage (%)
- Rural AI Usage (%)
- AI in Curriculum
- Gender Gap in AI Usage (%)